

Hotel Harmony – Booking Data EDA

Project Overview

This project analyzes hotel booking data to explore booking patterns, cancellations, guest behavior, room preferences, and average daily rates (ADR) across City Hotels and Resort Hotels.

Requirements

Python 3.x and Jupyter Notebook.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook.

Data Preparation

The original dataset contained 119,390 rows and 32 columns. After replacing placeholder NULL values, filling missing children values with zero, parsing reservation dates, and removing duplicate rows, 87,396 records remained.

Analysis Approach

Examined lead time, cancellations, arrival seasonality, special requests, guest countries, ADR by hotel type and market segment, distribution channels, previous cancellations, stay duration, room types, yearly ADR trends, parking needs, and relationships between booking changes and special requests. Visualizations included count plots, scatter plots, line charts, bar charts, and box plots.

Key Findings

The cleaned dataset contains 87,396 bookings, with 24,025 cancellations (27.5%). Average lead time is 79.89 days, and August is the most common arrival month. Portugal (PRT) is the most frequent guest country. City Hotel bookings have an average ADR of 110.99 compared with 99.03 for Resort Hotels; their cancellation rates are 30.0% and 23.5%, respectively. August has the highest total ADR among arrival months in this analysis. These findings describe this dataset and do not establish causation.

Challenges

The data includes placeholder NULL values and repeated rows that required handling. Booking records span different hotel types and market segments, making comparisons dependent on context. Advanced regression questions were excluded from the notebook's pandas/NumPy/Matplotlib-focused scope.

Recommendations

Review cancellation patterns by lead time, hotel type, and season when planning booking policies. Consider hotel- and segment-specific ADR trends when evaluating pricing. Further analysis could test cancellation drivers and guest composition using validated statistical models.

Conclusion

The project demonstrates how exploratory data analysis can reveal patterns in hotel bookings, cancellations, seasonality, and pricing. It provides a foundation for deeper analysis of revenue management and booking behavior.

Project Files

Hotel_Eda.ipynb – notebook containing the analysis; hotel_bookings.csv – source dataset.